

Library Management System – REST API Design to HATEOAS

PART 1 – JSON & XML

Books-schema.json

```
1. {
2.   "$schema": "https://json-schema.org/draft/2020-12/schema",
3.   "type": "array",
4.   "items":
5.     {
6.       "type": "object",
7.       "required": ["id", "title", "authors", "published_year", "available", "genres",
"publisher"],
8.       "properties":
9.         {
10.          "id":
11.            {
12.              "type": "integer",
13.              "minimum": 1
14.            },
15.          "title":
16.            {
17.              "type": "string"
18.            },
19.          "authors":
20.            {
21.              "type": "array",
22.              "minItems": 1,
23.              "items":
24.                {
25.                  "type": "string",
26.                  "minLength": 1,
27.                  "maxLength": 255
28.                }
29.            },
30.          "published_year":
31.            {
32.              "type" : "integer",
33.              "minimum": 1000,
34.              "maximum": 2026
35.            },
36.          "available":
37.            {
38.              "type": "boolean"
39.            },
40.          "genres" :
41.            {
42.              "type" : "array",
43.              "minItems": 2,
44.              "items":
45.                {
```

```

46.         "type": "string"
47.     }
48. },
49.     "publisher":
50.     {
51.         "type": "object",
52.         "properties":
53.         {
54.             "name":
55.             {
56.                 "type": "string"
57.             },
58.             "city":
59.             {
60.                 "type" : "string"
61.             }
62.         }
63.     }
64. }
65. }
66. }
67.

```

Books.json

```

1.  [
2.      {
3.          "id": 1,
4.          "title": "Harry Potter and the Sorcerer's Stone",
5.          "authors": ["J.K. Rowling"],
6.          "published_year": 1997,
7.          "available": true,
8.          "genres": ["Fantasy", "Adventure"],
9.          "publisher": {
10.             "name": "Bloomsbury",
11.             "city": "London"
12.          }
13.      },
14.      {
15.          "id": 2,
16.          "title": "The Hobbit",
17.          "authors": ["J.R.R. Tolkien"],
18.          "published_year": 1937,
19.          "available": false,
20.          "genres": ["Fantasy", "Classic"],
21.          "publisher": {
22.             "name": "George Allen & Unwin",
23.             "city": "London"
24.          }
25.      },
26.      {
27.          "id": 3,
28.          "title": "To Kill a Mockingbird",
29.          "authors": ["Harper Lee"],
30.          "published_year": 1960,
31.          "available": true,
32.          "genres": ["Fiction", "Classic"],
33.          "publisher": {
34.             "name": "J.B. Lippincott & Co.",
35.             "city": "Philadelphia"
36.          }
37.      },

```



```

32.             <xs:sequence>
33.                 <xs:element name="name" type="xs:string"/>
34.                 <xs:element name="city" type="xs:string"/>
35.             </xs:sequence>
36.         </xs:complexType>
37.     </xs:element>
38. </xs:sequence>
39. </xs:complexType>
40. </xs:element>
41. </xs:sequence>
42. </xs:complexType>
43. </xs:element>
44. </xs:schema>
45.

```

Books.xml

```

1. <?xml version="1.0" encoding="UTF-8"?>
2. <library xmlns="http://example.com/library">
3.     <book>
4.         <id>1</id>
5.         <title>Harry Potter and the Sorcerer's Stone</title>
6.         <authors>
7.             <author>J.K. Rowling</author>
8.         </authors>
9.         <published_year>1997</published_year>
10.        <available>true</available>
11.        <genres>
12.            <genre>Fantasy</genre>
13.            <genre>Adventure</genre>
14.        </genres>
15.        <publisher>
16.            <name>Bloomsbury</name>
17.            <city>London</city>
18.        </publisher>
19.    </book>
20.    <book>
21.        <id>2</id>
22.        <title>The Hobbit</title>
23.        <authors>
24.            <author>J.R.R Tolkien</author>
25.        </authors>
26.        <published_year>1937</published_year>
27.        <available>false</available>
28.        <genres>
29.            <genre>Fantasy</genre>
30.            <genre>Classic</genre>
31.        </genres>
32.        <publisher>
33.            <name>George Allen & Unwin</name>
34.            <city>London</city>
35.        </publisher>
36.    </book>
37.    <book>
38.        <id>3</id>
39.        <title>To Kill a Mockingbird</title>
40.        <authors>
41.            <author>Harper Lee</author>
42.        </authors>
43.        <published_year>1960</published_year>
44.        <available>true</available>
45.        <genres>

```

```

46.         <genre>Fiction</genre>
47.         <genre>Classic</genre>
48.     </genres>
49.     <publisher>
50.         <name>J.B. Lippincott & Co.</name>
51.         <city>Philadelphia</city>
52.     </publisher>
53. </book>
54. <book>
55.     <id>4</id>
56.     <title>1984</title>
57.     <authors>
58.         <author>George Orwell</author>
59.     </authors>
60.     <published_year>1949</published_year>
61.     <available>true</available>
62.     <genres>
63.         <genre>Dystopian</genre>
64.         <genre>Science Fiction</genre>
65.     </genres>
66.     <publisher>
67.         <name>Secker & Warburg</name>
68.         <city>London</city>
69.     </publisher>
70. </book>
71. <book>
72.     <id>5</id>
73.     <title>Pride and Prejudice</title>
74.     <authors>
75.         <author>Jane Austen</author>
76.     </authors>
77.     <published_year>1813</published_year>
78.     <available>false</available>
79.     <genres>
80.         <genre>Romance</genre>
81.         <genre>Classic</genre>
82.     </genres>
83.     <publisher>
84.         <name>T. Egerton</name>
85.         <city>London</city>
86.     </publisher>
87. </book>
88. </library>
89.

```

PART 2 – REST API Design

```

1. GET /books
2. GET /books/{id}
3. GET /books/{id}/authors
4. GET /book/{id}/authors/{id}
5. POST /books
6. PUT /books/{id}
7. PATCH /books/{id}
8. DELETE /books/{id}
9.
10. ?genre=romance&

```

```
11. ?rating=6&
12. ?author=Kurt%20Laja
13. ?title=The%20Notebook
```

PART 3 – HTTP Headers & Status Codes

```
1. GET endpoints
2.
3. GET /books
4. GET /books/{id}
5. GET /books/{id}/authors
6. GET /book/{id}/authors/{id}
7.
8. Content-Type: application/json
9. Authorization: Bearer <token>
10. Accept: application/json
11.
12. Status codes
13. - 200 OK
14. - 400 Bad Request
15. - 401 Unauthorized
16. - 403 Forbidden
17. - 404 Not Found
18. - 500 Internal Server Error
19.
20.
```

POST, PUT, PATCH, AND DELETE endpoints

```
1. POST /books
2. PUT /books/{id}
3. PATCH /books/{id}
4. DELETE /books/{id}
5.
6. POST:
7.   Headers:
8.   Content-Type: application/json
9.   Authorization: Bearer <token>
10.  Accept: application/json
11.
12.   Status Codes:
13.   - 200 OK
14.   - 201 Created
15.   - 400 Bad Request
16.   - 401 Unauthorized
17.   - 500 Internal Server Error
18.
19. PUT:
20.   Headers:
21.   Content-Type: application/json
22.   Authorization: Bearer <token>
23.   Accept: application/json
24.
25.   Status Codes:
```

```
26.     - 200 OK
27.     - 201 Created
28.     - 400 Bad Request
29.     - 401 Unauthorized
30.     - 500 Internal Server Error
31.
32. PATCH:
33.     Headers:
34.     Content-Type: application/json
35.     Authorization: Bearer <token>
36.     Accept: application/json
37.
38.     Status Codes:
39.     - 200 OK
40.     - 201 Created
41.     - 400 Bad Request
42.     - 401 Unauthorized
43.     - 500 Internal Server Error
44.
45. DELETE:
46.     Headers:
47.     Content-Type: application/json
48.     Authorization: Bearer <token>
49.     Accept: application/json
50.
51.     Status Codes:
52.     - 200 OK
53.     - 204 No Content
54.
```

PART 4 – HATEOAS

```
1. [
2.   {
3.     "id": 1,
4.     "title": "The Notebook",
5.     "year": 2001,
6.     "genre": "romance",
7.     "casts": ["Ryan Gosling", "Rachel McAdams"],
8.     "watched": true,
9.     "rating": 6.7,
10.    "_links": {
11.      "directors": {
12.        "href": "/movies/1/directors",
13.        "method": "GET"
14.      },
15.      "collection": {
16.        "href": "/movies",
17.        "method": "GET"
18.      },
19.      "reviews": {
20.        "href": "/movies/1/reviews",
21.        "method": "GET"
22.      },
23.      "watch": {
24.        "href": "/movies/1",
25.        "method": "GET"
26.      },
27.      "casts": {
```

```

28.         "href": "/movies/1/casts",
29.         "method": "GET"
30.     },
31.     "review": {
32.         "href": "/movies/1",
33.         "method": "POST"
34.     },
35.     "update": {
36.         "href": "/movies/1",
37.         "method": "PUT"
38.     },
39.     "delete": {
40.         "href": "/movies/1",
41.         "method": "DELETE"
42.     }
43. },
44. },
45. {
46.     "id": 2,
47.     "title": "Inception",
48.     "year": 2010,
49.     "genre": "science fiction",
50.     "casts": ["Leonardo DiCaprio", "Joseph Gordon-Levitt"],
51.     "watched": true,
52.     "rating": 8.8,
53.     "_links": {
54.         "directors": {
55.             "href": "/movies/2/directors",
56.             "method": "GET"
57.         },
58.         "collection": {
59.             "href": "/movies",
60.             "method": "GET"
61.         },
62.         "reviews": {
63.             "href": "/movies/2/reviews",
64.             "method": "GET"
65.         },
66.         "watch": {
67.             "href": "/movies/2",
68.             "method": "GET"
69.         },
70.         "casts": {
71.             "href": "/movies/2/casts",
72.             "method": "GET"
73.         },
74.         "review": {
75.             "href": "/movies/2",
76.             "method": "POST"
77.         },
78.         "update": {
79.             "href": "/movies/2",
80.             "method": "PUT"
81.         },
82.         "delete": {
83.             "href": "/movies/2",
84.             "method": "DELETE"
85.         }
86.     }
87. },
88. {
89.     "id": 3,
90.     "title": "The Godfather",
91.     "year": 1972,
92.     "genre": "crime",

```



```

93.     "casts": ["Marlon Brando", "Al Pacino"],
94.     "watched": false,
95.     "rating": 9.2,
96.     "_links": {
97.         "directors": {
98.             "href": "/movies/3/directors",
99.             "method": "GET"
100.        },
101.        "collection": {
102.            "href": "/movies",
103.            "method": "GET"
104.        },
105.        "reviews": {
106.            "href": "/movies/3/reviews",
107.            "method": "GET"
108.        },
109.        "watch": {
110.            "href": "/movies/3",
111.            "method": "GET"
112.        },
113.        "casts": {
114.            "href": "/movies/3/casts",
115.            "method": "GET"
116.        },
117.        "review": {
118.            "href": "/movies/3",
119.            "method": "POST"
120.        },
121.        "update": {
122.            "href": "/movies/3",
123.            "method": "PUT"
124.        },
125.        "delete": {
126.            "href": "/movies/3",
127.            "method": "DELETE"
128.        }
129.    }
130. },
131. {
132.     "id": 4,
133.     "title": "La La Land",
134.     "year": 2016,
135.     "genre": "musical",
136.     "casts": ["Emma Stone", "Ryan Gosling"],
137.     "watched": true,
138.     "rating": 8.0,
139.     "_links": {
140.         "directors": {
141.             "href": "/movies/4/directors",
142.             "method": "GET"
143.        },
144.        "collection": {
145.            "href": "/movies",
146.            "method": "GET"
147.        },
148.        "reviews": {
149.            "href": "/movies/4/reviews",
150.            "method": "GET"
151.        },
152.        "watch": {
153.            "href": "/movies/4",
154.            "method": "GET"
155.        },
156.        "casts": {
157.            "href": "/movies/4/casts",

```

```

158.         "method": "GET"
159.     },
160.     "review": {
161.         "href": "/movies/4",
162.         "method": "POST"
163.     },
164.     "update": {
165.         "href": "/movies/4",
166.         "method": "PUT"
167.     },
168.     "delete": {
169.         "href": "/movies/4",
170.         "method": "DELETE"
171.     }
172. },
173. },
174. {
175.     "id": 5,
176.     "title": "Parasite",
177.     "year": 2019,
178.     "genre": "thriller",
179.     "casts": ["Song Kang-ho", "Lee Sun-kyun"],
180.     "watched": false,
181.     "rating": 8.5,
182.     "_links": {
183.         "directors": {
184.             "href": "/movies/5/directors",
185.             "method": "GET"
186.         },
187.         "collection": {
188.             "href": "/movies",
189.             "method": "GET"
190.         },
191.         "reviews": {
192.             "href": "/movies/5/reviews",
193.             "method": "GET"
194.         },
195.         "watch": {
196.             "href": "/movies/5",
197.             "method": "GET"
198.         },
199.         "casts": {
200.             "href": "/movies/5/casts",
201.             "method": "GET"
202.         },
203.         "review": {
204.             "href": "/movies/5",
205.             "method": "POST"
206.         },
207.         "update": {
208.             "href": "/movies/5",
209.             "method": "PUT"
210.         },
211.         "delete": {
212.             "href": "/movies/5",
213.             "method": "DELETE"
214.         }
215.     }
216. },
217. ]
218.
219.
220.
221.

```